Manufacturing & Video-Analytics Intelligent Report

“I want the data from the mes application for machine utilization and from video analytics I want camera configuration information”

Generated: 2026-10-15 23:12:40
Time filter applied

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Executive Summary

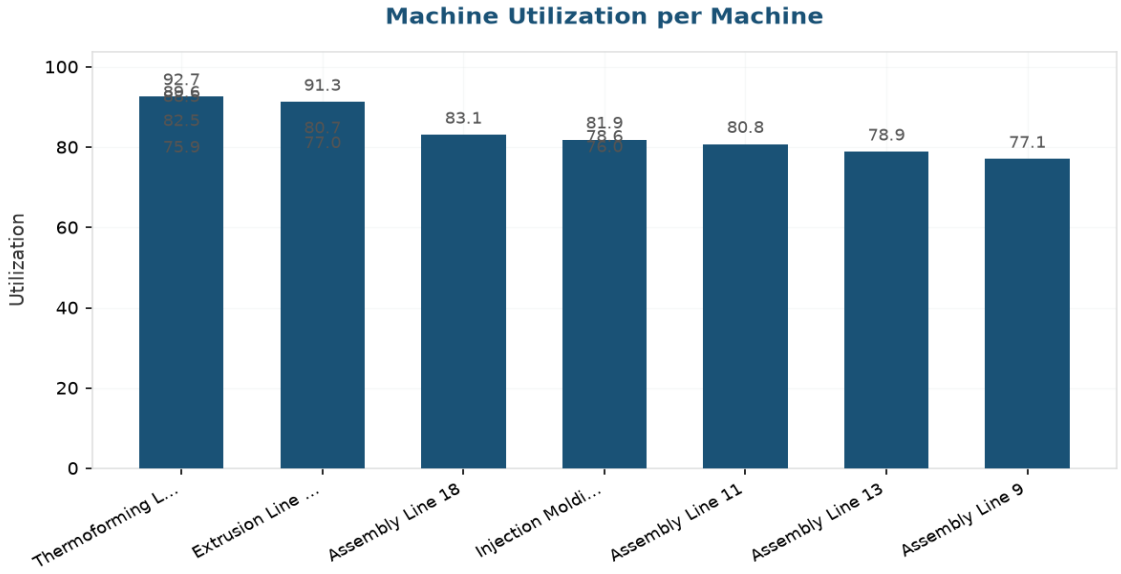
2	12	333	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, configuration, machine, utilization

Telemetry window: 2026-10-14T17:42:40Z → 2026-10-15T17:42:40Z (bucketed every 5m)

Machine Utilization Overview

The MES MachineMaster table contains 40 machines with a Utilization field ranging from a low of 43.4% to a high of 92.7%. The overall average utilization is 68.61%, indicating that on average the plant is operating below full capacity. While some machines are near full load (≈93%), a substantial number are below 60%, suggesting opportunities for load balancing or maintenance. Utilization values are unique per machine, reflecting individual performance rather than aggregated shifts.



Data preview — MachineMaster (showing 10 of 40 points/rows)

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
81.90	M001	1	Injection Molding Line 1	Injection Molding	772.96	178.64	1
45.00	M002	2	Thermoforming Line 2	Thermoforming	554.94	77.34	1
58.90	M003	3	Thermoforming Line 3	Thermoforming	826.22	229.64	1
75.90	M004	4	Thermoforming Line 4	Thermoforming	681.29	150.48	1
59.00	M005	5	Assembly Line 5	Assembly	120.46	235.09	1
69.70	M006	6	Injection Molding Line 6	Injection Molding	190.81	333.32	1
43.40	M007	7	Extrusion Line 7	Extrusion	705.28	335.53	1
56.50	M008	8	Thermoforming Line 8	Thermoforming	572.02	66.41	1
77.10	M009	9	Assembly Line 9	Assembly	283.00	388.73	1
70.40	M010	10	Blow Molding Line 10	Blow Molding	943.44	217.79	1

Shift■Level Utilization Summary

The CapacityAnalysis table records 50 utilization entries for 4 machines across 4 shifts. UtilizationPercent varies between 37.5% and 100%, with an overall average of 72.92%. This average is higher than the MachineMaster overall average, indicating that certain shifts achieve near■full capacity while others lag. The data is complete and shows no outliers, but without time stamps we cannot plot trends.

Data preview — CapacityAnalysis (showing 10 of 50 points/rows)

MachineId	UtilizationPercent	CapacityId	CustomerId	ShiftId
1	100.00	1	1	4
1	51.62	2	1	1
1	52.62	3	1	4
1	100.00	4	1	3
1	100.00	5	1	2
1	61.88	6	1	2
1	67.25	7	1	1
1	59.75	8	1	3
1	64.00	9	1	4
1	76.38	10	1	1

Camera Configuration Details

Three cameras are defined in the video analytics system. Camera 1 (Flarehub) and Camera 2 (Luxsphere) have no department, location, or plant assignments, while Camera 6 (Entry Gate) is linked to department 1, location 1, and plant 1. All three cameras report a status of "online" and are managed by the user "admin".

Data preview — cameras (showing 3 of 3 points/rows)

camera_number	department_id	location_id	name	plant_id	status	user_id
1	None	None	Flarehub	None	online	admin
2	None	None	Luxsphere	None	online	admin
6	1	1	Entry Gate	1	online	admin

Camera Status Log Discrepancy

The camera_status_logs measurement contains 50 records, all reporting the status "offline" for camera IDs 1, 2, and 3. This contradicts the cameras table where all three cameras are listed as "online". The logs are complete (100% completeness) but appear to capture only one status value, suggesting a possible data collection or configuration issue that should be investigated.

Data preview — camera_status_logs (showing 10 of 50 points/rows)

camera_id	status
1	offline
1	offline
2	offline
3	offline
1	offline
2	offline
3	offline
1	offline
2	offline
3	offline

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	71.4%	0	Needs review
camera_status_logs	50	100.0%	0	Good
MachineMaster	40	100.0%	0	Good
CapacityAnalysis	50	100.0%	0	Good
Machine	40	100.0%	0	Good
MachineFGMapping	50	100.0%	0	Good
MachineMaterialMapping	50	100.0%	0	Good
WeighingMachine	50	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
groq/openai/gpt-oss-120b	4,435	1,419	5,854	N/A